

Timetable Generator

UI Design Document

Screens, Navigation and User Flow of the Admin & Faculty Web Application

Submitted for Project Viva • September 2026

1. What This Document Covers

“UI design” here means the plan behind how a person actually uses the application: which screens exist, what each one lets a user do, how a user moves from one screen to the next, and why the screens are organised the way they are for the different roles (admin, HOD, teacher, student) that use this system. It is documented after the fact from the working application rather than drawn up front, since the interface already exists as working React code — this document explains and justifies that interface rather than proposing a new one.

2. Design Goals

Guided setup over free-form data entry. A brand-new college has never used the system before and doesn't know what data it needs. A fixed step-by-step wizard (years/sections → subjects → curriculum → electives → lab batches → teachers → rooms → import) removes the guesswork about ordering.

No numeric weights in the interface. Soft scheduling preferences are set as plain-language importance levels (e.g. “Very Important”), not raw numbers a non-technical administrator would have to guess at.

Fail loud, fail specific. Wherever generation can fail (a missing lab batch, an overloaded teacher), the UI is designed to surface the exact named cause from the backend's diagnostics rather than a generic error banner.

One screen, one role's job. Route-level access control means an HOD only ever sees their own department's runs, and a teacher or student only ever lands on their personal timetable — nobody sees a control they cannot use.

The generated result is always reviewable before it's final. A newly generated timetable is inspected in the viewer, can be edited slot-by-slot, and is only published when the admin is satisfied — generation and publishing are always two separate steps.

3. Information Architecture & Role-Based Navigation

Every route in the application is protected and role-gated at the router level (`frontend/src/App.tsx`). On login, a user is sent straight to the area relevant to their role instead of a generic landing page:

Role	Lands on	Can also reach
Admin	Dashboard (setup checklist)	Setup Wizard, Scheduling Rules, Settings, Danger Zone, Timetable Runs,
HOD	Timetable Runs (own department)	Timetable Viewer, Version History — scoped to their department or
Faculty	My Timetable	Their own personal weekly schedule, exportable as .ics
Student	My Timetable	Their own section's published schedule, exportable as .ics

3.1 Setup Wizard Sub-Navigation (Admin only)

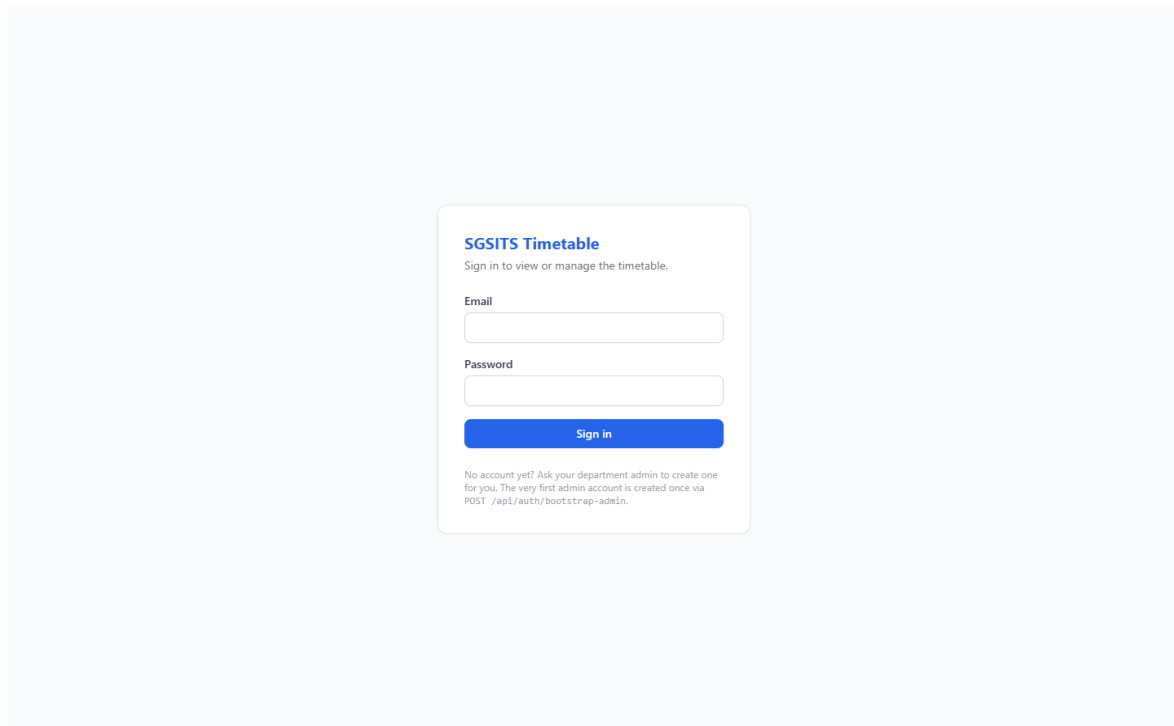


1. Years & Sections → 2. Subjects → 3. Curriculum → 4. Electives → 5. Lab Batches → 6. Teachers → 7. Rooms → 8. Bulk Import

Each step is its own route under `/setup/*`, so an admin can leave the wizard midway and return to exactly the step they were on. Bulk Import is an alternative to filling in steps 1–7 by hand — it accepts a spreadsheet with one sheet per entity and a preview step that validates without writing anything.

4. Screen-by-Screen Walkthrough

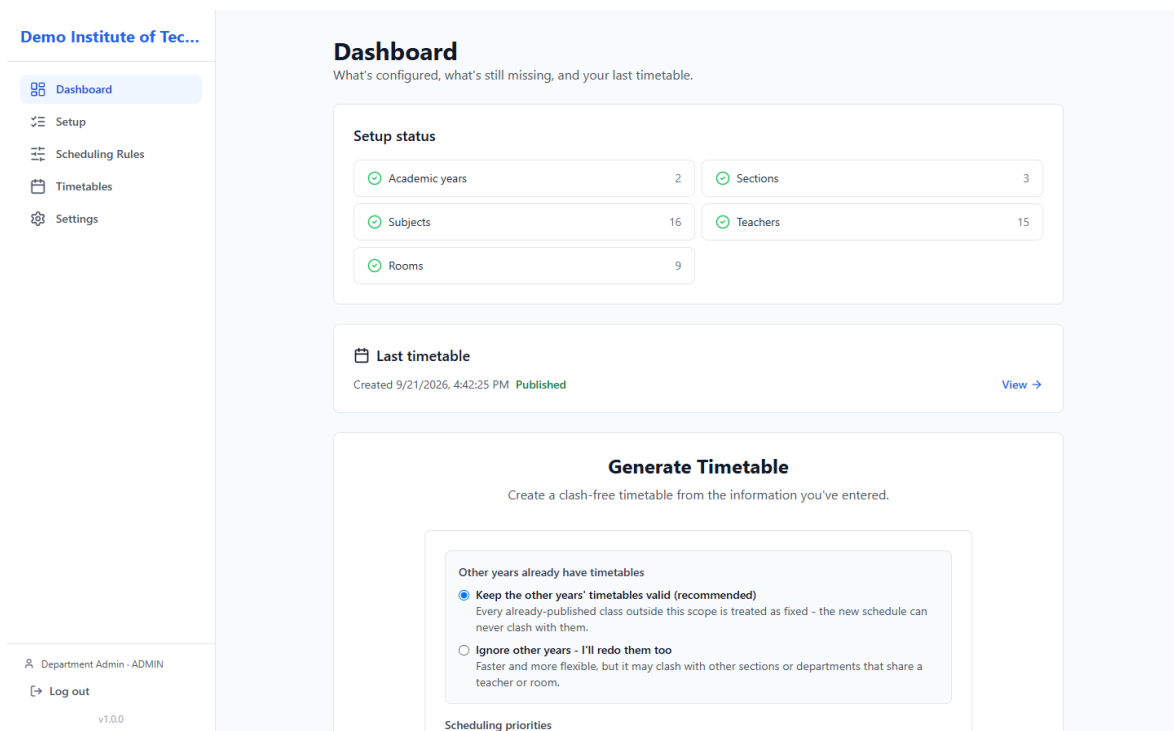
4.1 Login



Login screen — email/password, role-aware redirect after sign-in.

A single email/password form. On success, the user is redirected based on role as described in Section 3, so an admin and a student never see the same first screen.

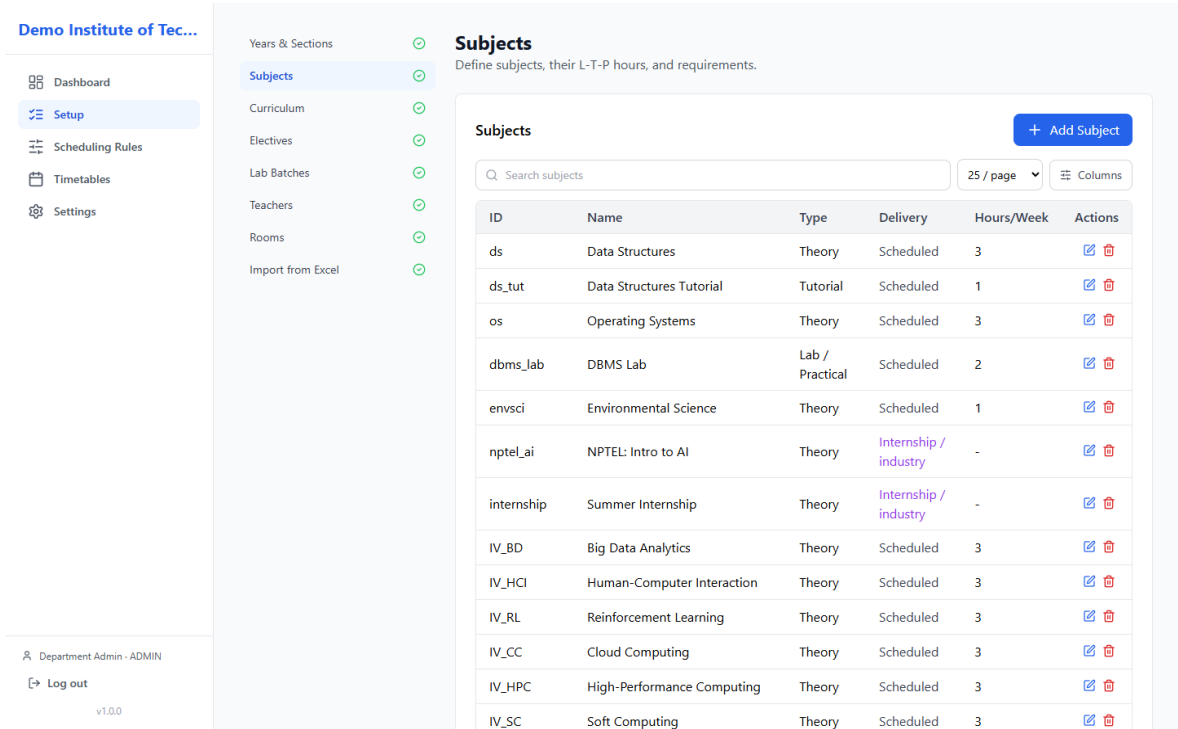
4.2 Dashboard (Admin)



Admin dashboard — setup checklist and the Generate action.

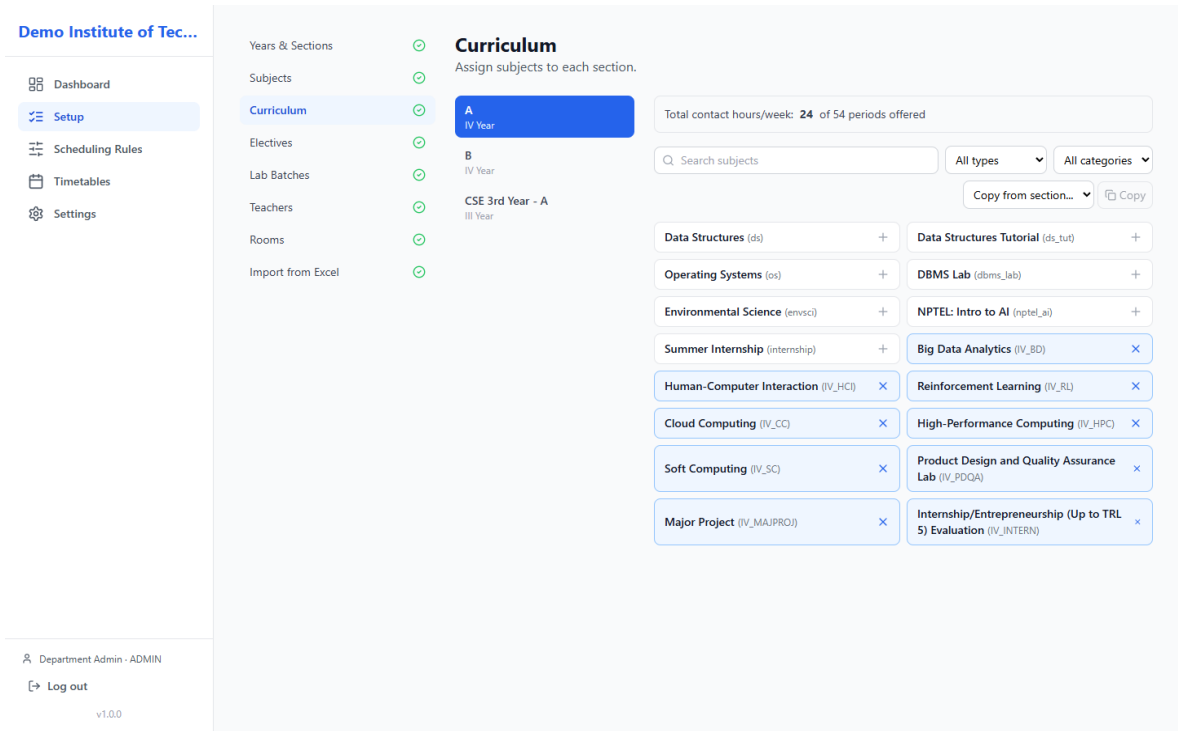
The admin's home screen is a checklist of setup prerequisites (department created? subjects added? teachers mapped? rooms defined?) rather than a blank dashboard, so a first-time user always knows the next concrete step instead of hunting through a menu.

4.3 Setup Wizard — Subjects & Curriculum



Subjects step — L-T-P hours, type and delivery mode per subject.

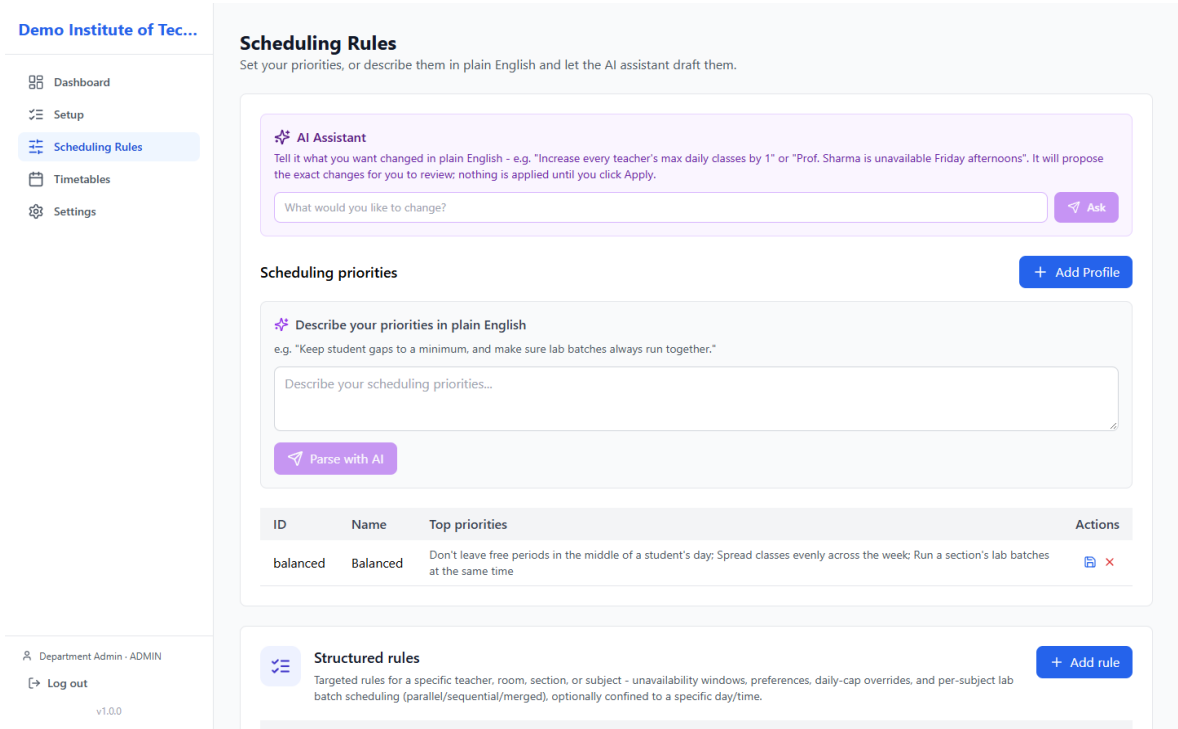
Each subject is entered with its L-T-P hours, type (theory/lab/tutorial) and delivery mode. Subjects marked “Internship / industry” — visible here as “NPTEL: Intro to AI” and “Summer Internship” — are shown with 0 scheduled hours and are excluded from the generated grid entirely, while still being tracked for credit purposes elsewhere in the system.



Curriculum step — assigning subjects to a specific section.

The Curriculum step is where a subject is actually attached to a section; a subject that exists but isn't assigned to any section here is simply never scheduled, which keeps subject definitions reusable across multiple sections and years.

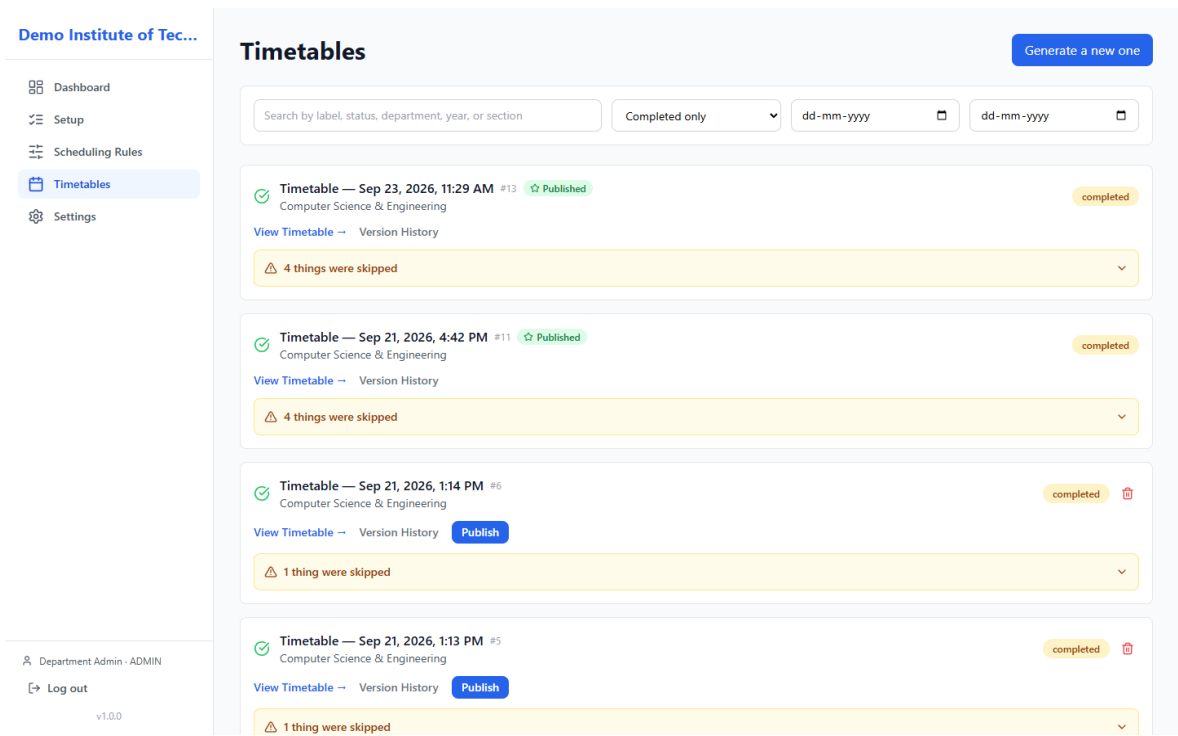
4.4 Scheduling Rules



Scheduling Rules — priorities in plain English, plus an AI assistant and structured per-teacher rules.

This is the screen behind the “no numeric weights” design goal from Section 2. An admin either types priorities as a sentence (“Keep student gaps to a minimum, and make sure lab batches always run together”) which gets parsed into the rule catalogue, or works from a named profile such as “Balanced”, whose top priorities are shown back in plain English rather than as coefficients. A separate **Structured rules** panel below it holds targeted, factual overrides — a specific teacher's unavailability window, a per-subject lab batching mode — that don't make sense as a vague priority. The AI assistant here only ever *proposes* a change; nothing is written until the admin clicks Apply.

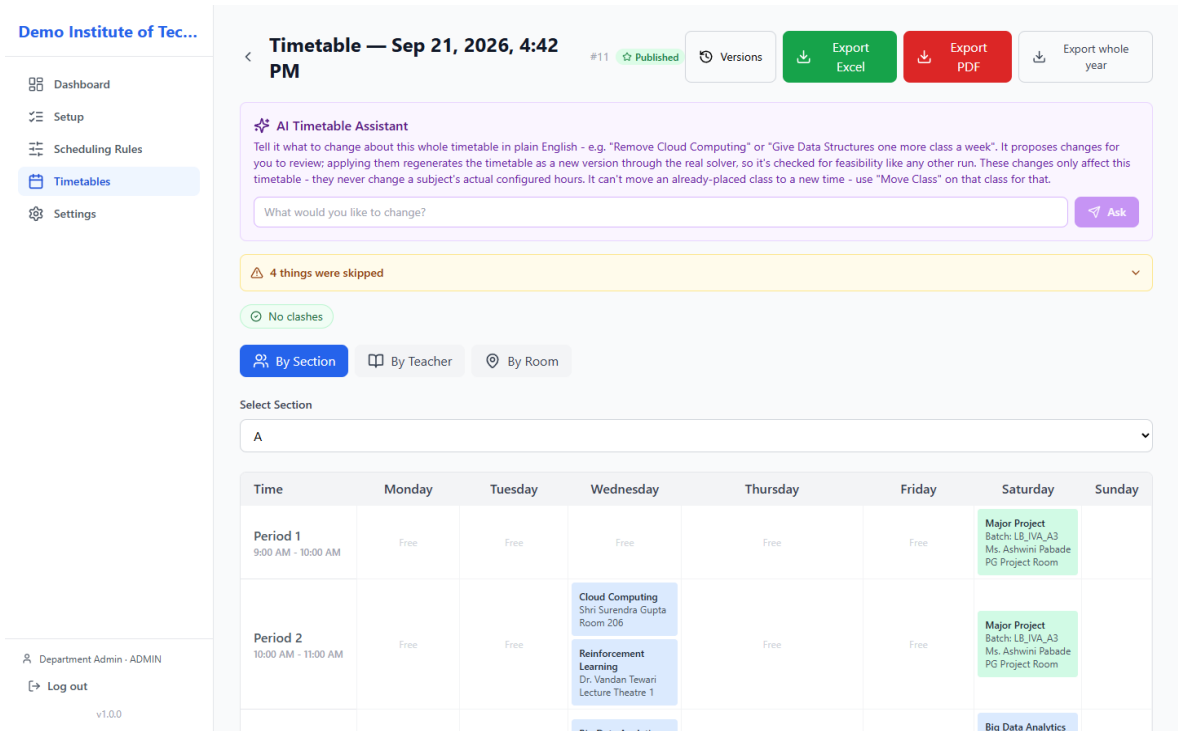
4.5 Timetable Runs



Timetables list — every generation attempt, its status, and publish/delete actions.

A list of every generation attempt for the department, each with its status, scope, and any “N things were skipped” warnings expandable in place. Two different runs can be published at once as long as their scopes don't overlap (e.g. one run per academic year), which is exactly how the screenshot in Section 4.6 and the earlier dashboard screenshot were produced — run #11 for the IV-Year sections, run #13 for III-Year CSE-3A, both live simultaneously.

4.6 Timetable Viewer



Timetable Viewer — the generated weekly grid, with a live clash badge, export options and the AI Timetable Assistant.

The core screen of the product: a weekly grid of the generated schedule, switchable between By Section / By Teacher / By Room, with a “No clashes” badge confirming the independent validator from the pipeline

document passed. From here an admin can compare against a previous version, export to Excel/PDF or the whole year at once, or describe a change in plain English to the AI Timetable Assistant — which always re-solves through the real solver rather than editing the grid directly, so any accepted change is itself re-validated like any other run.

4.7 My Timetable (Faculty and Student views)

Demo Institute of Tec...

My Timetable

My Timetable

Your class schedule.

Add to my calendar

Day	Period 1 9:00 AM - 10:00 AM	Period 2 10:00 AM - 11:00 AM	Period 3 11:00 AM - 12:00 PM	Period 4 12:00 PM - 1:00 PM	Period 5 1:00 PM - 2:00 PM	Period 6 2:00 PM - 3:00 PM	Period 7 3:00 PM - 4:00 PM	Period 8 4:00 PM - 5:00 PM	Period 9 5:00 PM - 6:00 PM
Monday	—	—	—	—	os It1	envsci R_TUT	—	—	—
Tuesday	—	—	—	—	os R_C4	ds R_PG	—	—	—
Wednesday	—	—	—	—	—	—	—	dbms_lab Batch cse3a_b1 lab1 dbms_lab Batch cse3a_b2 lab2	—
Thursday	—	dbms_lab Batch cse3a_b1 lab1 dbms_lab Batch cse3a_b2 lab2	—	—	—	—	—	—	—
Friday	—	—	—	—	ds It1	ds_tut R_206	—	—	—
Saturday	—	—	—	—	—	—	ds R_PG	os R_PG	—
Sunday	—	—	—	—	—	—	—	—	—

A Student - STUDENT

Log out

v1.0.0

My Timetable — a student's own read-only weekly schedule.

Demo Institute of Tec...

My Timetable

My Timetable

Your teaching schedule.

Add to my calendar

Day	Period 1 9:00 AM - 10:00 AM	Period 2 10:00 AM - 11:00 AM	Period 3 11:00 AM - 12:00 PM	Period 4 12:00 PM - 1:00 PM	Period 5 1:00 PM - 2:00 PM	Period 6 2:00 PM - 3:00 PM	Period 7 3:00 PM - 4:00 PM	Period 8 4:00 PM - 5:00 PM	Period 9 5:00 PM - 6:00 PM
Monday	—	—	—	—	—	envsci R_TUT	—	—	—
Tuesday	—	—	—	—	—	ds R_PG	—	—	—
Wednesday	—	—	—	—	—	—	—	dbms_lab Batch cse3a_b1 lab1	—
Thursday	—	dbms_lab Batch cse3a_b1 lab1	—	—	—	—	—	—	—
Friday	—	—	—	—	ds It1	ds_tut R_206	—	—	—
Saturday	—	—	—	—	—	—	ds R_PG	—	—
Sunday	—	—	—	—	—	—	—	—	—

Prof. Sharma - FACULTY

Log out

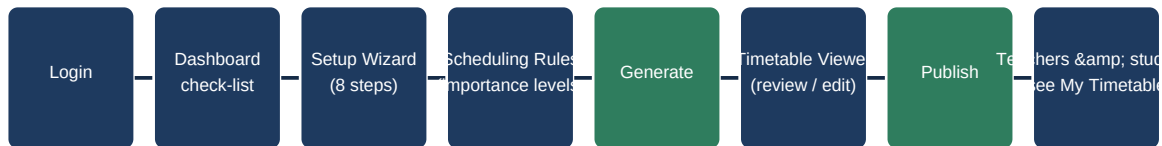
v1.0.0

My Timetable — a faculty member's own teaching schedule, scoped to only their classes.

Both roles land on the same route, but the query is scoped server-side to the logged-in user: a student sees every class their section attends, while a teacher sees only the classes they personally teach across whatever sections they're assigned to. A one-click “Add to my calendar” export lets either of them

subscribe to it from their own calendar app. Nothing on this screen is editable — it deliberately mirrors what is currently published, nothing more.

5. End-to-End User Flow (Admin, First-Time Setup)



Generate and Publish (in green) are kept as two separate, deliberate actions rather than one step, so a newly generated schedule can always be reviewed — and, if necessary, edited or discarded — before it becomes visible to teachers and students.

6. Conclusion

The interface is organised around two ideas: get an administrator to a correct, complete dataset as quickly as possible through a guided wizard, and never let an automatically generated result reach a teacher or student without a human review step in between. Role-based routing keeps every screen relevant to the person viewing it, and the same diagnostic detail the solver produces on failure is surfaced directly in the UI rather than flattened into a generic error.